

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2020 - 2021 (Even Semester)
Innovative Practices Description

Unit I Drive Characteristics

Degree, Semester & Branch	: VI Semester B.E. EEE
Course Code & Title	: EE8601 Solid State Drives
Name of the Faculty member	: Mrs. S. Jeyanthi
Name of the Topic	: Typical load torque characteristics
Name of the Innovative Practice	: Mini Map
Date & Duration	: 01.03.2021& 15 Minutes

Description:

A mini map is a diagram used to visually organize information. A mini map is hierarchical and shows relationships among pieces of the whole. Mini mapping in a learning environment which requires an enhanced way of thinking, learning and presenting. Mini Mapping enhances the learning capacity in terms of number of ideas generated as well as improves presentation focus. Mini maps are used to generate, visualize, structure and classify ideas. Mini maps go under variety of names. They are also known as concept maps, semantic mapping, knowledge mapping, think links, graphic organizers or cognitive maps.

Learning Outcome:

- The students will be able to visualize the concepts and organize information.

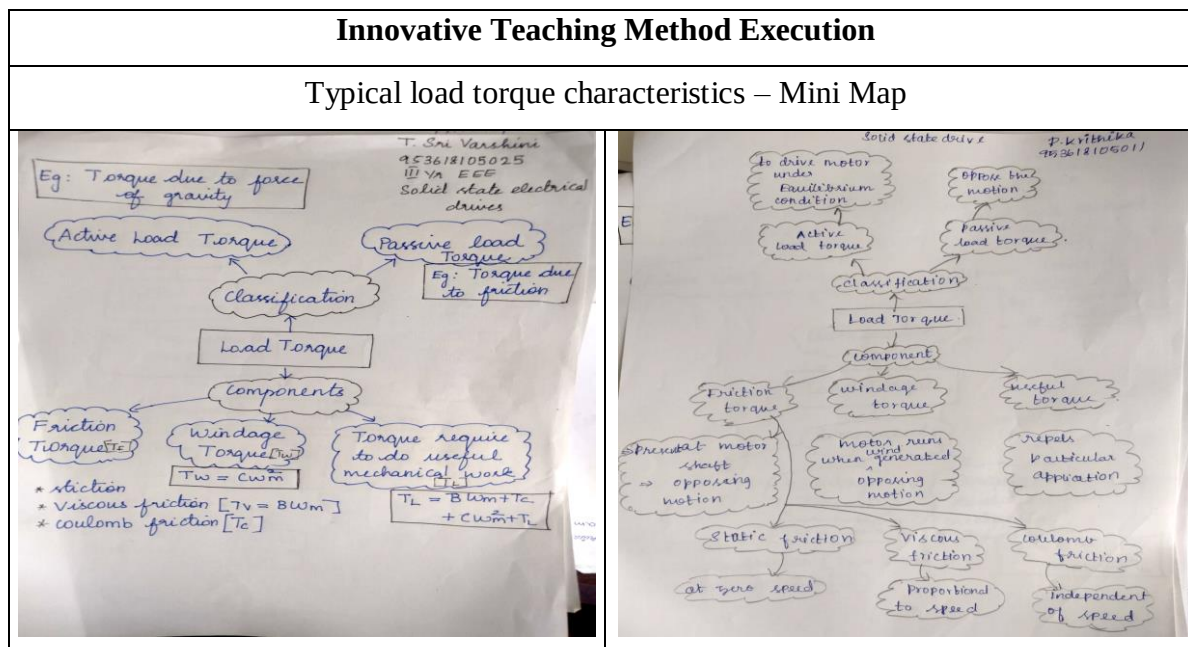
Justification for choosing the topic using Mini Map activity:

Mini mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use mini mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches.

Implementation (Plan & Execution) with Proof:

Mini mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a mini map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

In my subject Mini map was conducted for Unit-I Drive characteristics for the topic typical load torque characteristics. The students are voluntarily drawn the map. The sample copy of the mini map was done by the students was shown below.



Reflective Critique:

Benefits:

The benefits of Mini Maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mini map made slow learners to remember the information more quickly.

Challenges:

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to draw the map to indicate relationships between the topics in typical load torque characteristics. I planned the activity for 10 minutes only. But in real scenario it takes 15 minutes to complete the whole topics in Unit-I.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	1	1	-	-	1	2	1	-	1

Reference:

1. GopalK.Dubey, Fundamentals of Electrical Drives, Narosa Publishing House, 1992.

CO1: After completion of the course, it is expected that students will be able to explain the steady state operation and transient dynamics of a motor load system.

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Innovative Practices Description

Unit II Converter / Chopper Fed DC Motor Drive

Degree, Semester & Branch	: VI Semester B.E. EEE
Course Code & Title	: EE8601 Solid State Drives
Name of the Faculty member	: Mrs. S. Jeyanthi
Name of the Topic	: Applications of DC motor drives
Name of the Innovative Practice	: One Minute Paper
Date & Duration	: 15.03.2021&05 Minutes

Description:

Minute paper is a short writing activity within the classroom in which students generate answers in response to questions asked by the teacher regarding the concepts taught in class. These questions stimulate a student to reflect on the lesson taught and learned and provides feedback regarding their understanding which helps the teachers to plan ahead for the next class. Its major advantage is that it provides rapid feedback on whether the instructor's main idea and what the students perceived as the main idea are the same.

Learning Outcome:

- The students will be able explain the applications of DC motor drive in various industries.

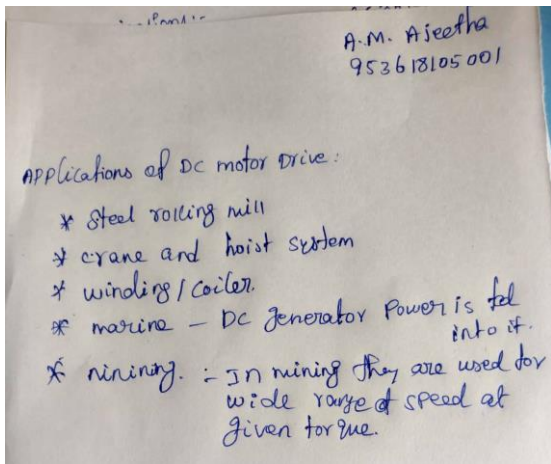
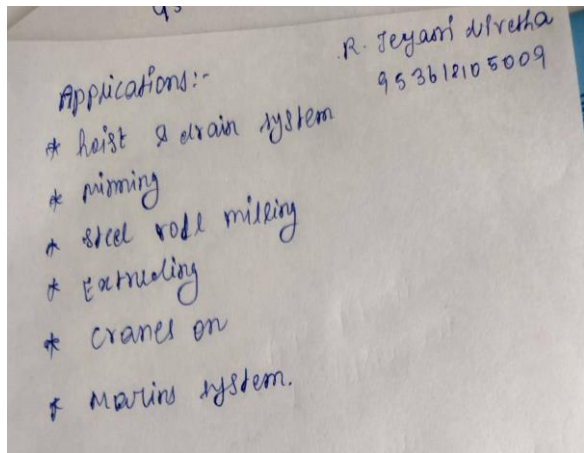
Justification for choosing the topic using one minute paper activity:

Applications of dc motor drive in wide range of industries due to their constant speed operation, dynamic response and high starting current. Since, one minute paper activity provides a conceptual bridge between successive class periods. Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

Implementation (Plan & Execution) with Proof:

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in

that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

Innovative Teaching Method Execution	
Applications of DC motor drive– One minute paper	
	

Reflective Critique:

1. Pre-implementation Reflection :

• Benefits:

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

• Challenges:

- In the class mostly boys hesitate to answer the questions.
- Time utilization for conducting activity.

2. Post-implementation Reflection :

• Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

• Challenges:

- Slow learners were not able to understand some topics during discussion hours.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	2	1	1	1	-	-	1	2	1	-	1

References:

1. VedamSubramanyam, “ Electric Drives Concepts and Applications ”, 2e, McGraw Hill, 2016
2. N.K. De., P.K. SEN” Electric drives” PHI, 2012.

CO2: After completion of the course, it is expected that students will be able to examine the operation of converter/chopper fed dc drive.